# Canonical UBT Action Audit for GAP-10T-DYN and GAP-10D: Fixed-Background Spin Current, Lorentz Pairing Rigidity, and Affine No-Go

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**Abstract**

We audit whether the canonical UBT action, as currently written in the repository, derives the Cartan torsion equation and the Einstein–$\Lambda$ tetrad equation without importing them. The answer is negative and is now made precise. We classify every term of the working action into derived, assumed, imported, or undefined. In the effective Palatini variation, with tetrad, metric, volume form and $\Theta$ held fixed while the independent Lorentz connection varies, we derive exactly the tree-level matter spin current of the kinetic term for the pure-pair representative $A_\mu = \Omega_\mu$, $B_\mu = -\Omega_\mu^\ddagger$. We prove that it couples only to the anti-Hermitian part of $\Theta$ whenever $D_\mu\Theta$ lies in the Lorentz slice, and prove pointwise rigidity for a nondegenerate standard tetrad. Consequently the fixed-background minimal Hilbert–Palatini + kinetic branch forces nonzero torsion on the complement of at most one point of every flat affine representer. We further classify all real symmetric Lorentz-invariant bilinear forms on the Lorentz slice: they are scalar multiples of the Minkowski form realized by the $\sharp$ pairing. The Hilbert–Schmidt $\ddagger$ pairing is Euclidean on the slice and fails boost invariance. Thus no nonzero nondegenerate Lorentz-invariant pairing choice removes the affine obstruction. The full composite $\Theta$-only variation, in which $e$, $g$ and the connection are functions of $D\Theta$, is not computed here. GAP-10T-DYN and GAP-10D remain NARROWED with the remaining lemmas named explicitly. All algebraic statements are verified by `tools/verify_canonical_spin_current.py`.

## 1 Canonical field content

The single fundamental field is $\Theta(q,\tau) \in \mathbb{B} \simeq \mathrm{Mat}(2,\mathbb{C})$. The involution $\ddagger$ is Hermitian conjugation; $\sharp$ is the $2\times 2$ adjugate, $X^\sharp = \mathrm{tr}(X)\mathbf{1} - X$. The classical Lorentz slice is $W_L = \{X : X^\ddagger = -X\}$ with basis $E_0 = i\mathbf{1}$, $E_k = -i\sigma_k$, and the central Jordan identity $\frac{1}{2}(X^\sharp Y + Y^\sharp X) = \eta(X,Y)\mathbf{1}$ holds on $W_L$ [verified: C1]. The covariant jet is $E_\mu = \mathcal{N}_0^{-1/2} D_\mu\Theta$ with

$$D_\mu\Theta = \partial_\mu\Theta + \Omega_\mu\Theta + \Theta\,\Omega_\mu^\ddagger, \qquad \Omega_\mu \in \mathfrak{sl}(2,\mathbb{C}), \tag{1}$$

which is the unique no-new-field representative established by GAP-10I-PAIR-KIN. The connection action preserves $W_L$ [C1].

## 2 Existing action

The working action in `canonical/THEORY/canonical/canonical_action.tex` is

$$S \;=\; S_{\mathrm{HP}}[e,\omega] \;+\; S_\Theta, \qquad S_\Theta = \tfrac{1}{2}\int \sqrt{-g}\,\langle D_\mu\Theta,\, D^\mu\Theta\rangle - \kappa V[\Theta], \tag{2}$$

with $S_{\mathrm{HP}}$ the Hilbert–Palatini/Einstein–Cartan first-order term with cosmological constant. Two pairings occur in the corpus and both are audited algebraically: the $\ddagger$ Hilbert–Schmidt pairing $\langle X, Y\rangle_{\ddagger} = \mathrm{Re}\,\mathrm{tr}(X^{\ddagger}Y)$ and the $\sharp$ scalar pairing $\langle X, Y\rangle_{\sharp} = \mathrm{Re}\,\mathrm{Sc}(X^{\sharp}Y)$. Only the latter is invariant under the full Lorentz action on $W_L$; the former is retained as a corpus comparison and is invariant only under the rotation subgroup on that slice (Theorem 3).

### Dependency classification (audit item 2)

| Object | Class | Basis |
|---|---|---|
| $\Theta$, $\mathbb{B}$, $\ddagger$, $\sharp$, $W_L$ | (a) derived/definitional | core algebra files |
| $E_\mu = \mathcal{N}_0^{-1/2} D_\mu\Theta$ | (a) derived | covariant-tetrad chain |
| $g_{\mu\nu}$ from central Jordan identity | (a) derived | rank theorem [L1] |
| $\Omega_\mu$ pure-pair form (1) | (a) derived | GAP-10I-PAIR-KIN [L1] |
| $S_\Theta$ kinetic form and normalization | (b) low-energy assumption | form assumed; Lorentz covariance selects $\sharp$ on $W_L$ up to scale, but sign/normalization are not fixed |
| Spin current $\tau^{\mu}{}_{ab}$ | (a) **derived here** | Theorem 1 |
| $S_{\mathrm{HP}}$, $\Lambda$ term | (c) standard import | stated as effective branch |
| $\kappa$, sign and value | (d) undefined from UBT | § Coefficient matching |
| Matter stress tensor $T_{\mu\nu}$ | (a) derived given (2) | `canonical/t_munu/` |
| Potential $V[\Theta]$ | (b) assumption | unconstrained by this audit |

## 3 Independent versus composite variations

Two inequivalent schemes exist and must not be conflated. *Independent (Palatini) scheme:* $e$ and $\omega$ vary independently. For the connection variation performed below, $e$, $g$, $\sqrt{-g}$, $\Theta$ and the index-raising operation are held fixed. In that variation $S_\Theta$ depends on $\omega$ only algebraically — the kinetic density is a polynomial of total degree two in the components of $\Omega_\mu$ with no derivatives of $\Omega_\mu$ [C2]. Hence the $\omega$-variation of $S_\Theta$ is an algebraic matter current and *no curvature invariant arises from this fixed-background kinetic term at tree level*; every curvature term in the effective field equations originates in the imported $S_{\mathrm{HP}}$.

*Composite scheme:* if the defining relation $e_\mu{}^a \sim D_\mu\Theta$ and $\omega = \mathring{\omega}(e) + K$ is imposed before variation, then $g$, $\sqrt{-g}$, index raising and the composite connection all vary with $\Theta$. The current formula below is then only the direct $\delta_\omega D\Theta$ contribution, not the full Euler–Lagrange variation. The induced quadratic form in $\partial e$ is $\Theta$-dependent and is *not* shown to reproduce the Einstein–Hilbert contraction. Deciding this full composite variation is part of the named remaining lemma below, not a result of this audit.

## 4 Torsion equation

In the independent scheme, $\delta S/\delta\omega$ yields the Cartan equation $\epsilon_{abcd}T^a \wedge e^b = \kappa\,\tau_{cd}$ with the $24\times 24$ torsion map pointwise invertible for a nondegenerate tetrad (GAP-10T-PALATINI [L1]). The new content is the source itself.

**Theorem 1** (Exact fixed-background tree-level spin current)**.** *For the pure-pair representative (1), holding $e$, $g$, $\sqrt{-g}$ and $\Theta$ fixed, the variation $\Omega_\mu \to \Omega_\mu + \varepsilon M$, $M \in \mathfrak{sl}(2,\mathbb{C})$, gives for either audited pairing*

$$\tau^\mu(M) \;=\; \big\langle\, M\Theta + \Theta M^{\ddagger},\; D^\mu\Theta \,\big\rangle. \tag{3}$$

**Lemma 1** (Slice lemma)**.** *If $D^\mu\Theta \in W_L$ (as required by the tetrad construction), then $\tau^\mu(M)$ depends only on the anti-Hermitian part of $\Theta$; the Hermitian part decouples identically. Verified exactly for both pairings [C3].*

**Theorem 2** (Pointwise rigidity). *Let $D_\mu\Theta = \sqrt{\mathcal{N}_0}\, E_\mu$ with $\{E_\mu\}$ the standard nondegenerate slice basis. Then $\tau^\mu(M) = 0$ for all $\mu$ and all $M \in \mathfrak{sl}(2,\mathbb{C})$ iff the anti-Hermitian part of $\Theta$ vanishes at that point. Verified exactly for both pairings [C4].*

**Corollary 1** (Flat affine no-go). *On every affine representer $\Theta = \Theta_0 + \sqrt{\mathcal{N}_0}\, E_\nu x^\nu$ (GAP-10I-SR), the anti-Hermitian part of $\Theta(x)$ is an affine non-constant $W_L$-valued map and vanishes at most at one point. Hence $\tau \neq 0$ on the complement of at most one point; the $x$-gradient of every component of $\tau$ is a $\Theta_0$-independent constant ($\pm 2\mathcal{N}_0$ for $\ddagger$, $\pm\mathcal{N}_0$ for $\sharp$) [C5]. By invertibility of the Cartan map, the minimal branch* (2) *forces nonzero torsion there, so the flat inertial, torsion-free solution is* not *a solution of the minimal branch.*

**Theorem 3** (Lorentz-invariant pairing rigidity). *Let $B$ be a real symmetric bilinear form on $W_L$ and require infinitesimal invariance under the full $\mathfrak{sl}(2,\mathbb{C})$ action $\delta_M X = MX + XM^\ddagger$. In the basis $(i\mathbf{1}, -i\sigma_1, -i\sigma_2, -i\sigma_3)$, the equations $J_M^T B + BJ_M = 0$ for the six Lorentz generators have the one-dimensional solution space*

$$B = c\ \mathrm{diag}(-1, 1, 1, 1).$$

*The $\sharp$ scalar pairing realizes the representative with $c = 1$. The $\ddagger$ Hilbert–Schmidt pairing restricts instead to $2\mathbf{1}_4$; it is rotation invariant but fails all three boost-invariance checks [C6]. Consequently every nonzero nondegenerate symmetric Lorentz-invariant pairing is proportional to the $\sharp$ pairing and inherits Corollary 1. Pairing selection alone cannot annihilate the affine current without becoming degenerate/zero or breaking Lorentz invariance.*

This is an exact flat-space counterpart of the curved Gaussian-patch scaling obstruction ($K \sim 1/\rho$ kinematic versus $\tau \sim \rho$ sourced): both identify the same structural mismatch of the minimal branch, by independent arguments.

## 5 Tetrad equation

In the independent scheme, $\delta S/\delta e$ yields $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ with $T_{\mu\nu}$ the canonical stress tensor of $S_\Theta$; the Einstein tensor originates entirely in $S_{\mathrm{HP}}$ [C2]. No derivation of the tetrad equation from $S_\Theta$ alone exists in the corpus, and this audit proves none.

## 6 Coefficient matching

Neither the sign nor the magnitude of $1/4\kappa$, nor $\Lambda$, is fixed by $\mathcal{N}_0$, by the $\Theta$ vacuum, or by any repository derivation. The only induced-gravity infrastructure present (`scripts/spectral/heat_kernel_trace`
computes flat-torus spectral traces and does not reach the curved Seeley–DeWitt $a_2$ coefficient. The coefficient therefore remains an explicit open subgap (part of GAP-10D).

## 7 No-circularity audit

(i) The spin current (3) is derived from $S_\Theta$ alone; no Einstein or Cartan equation is substituted anywhere in Theorems 1–3. (ii) The Cartan equation is used only *after* the current is derived, and only its proved algebraic invertibility [L1] is invoked. (iii) No step assumes the conclusion of GAP-10D; the Einstein–$\Lambda$ endpoint is cited exclusively as a property of the imported $S_{\mathrm{HP}}$. (iv) No new independent fields are introduced; the audit works entirely in the pure-pair representative.

## 8 Exact closure verdict

Verdict

**GAP-10T-DYN: NARROWED** (not closed, not closable in the minimal branch as written). Newly derived inside it: the exact fixed-background tree-level spin current (Theorem 1, conditional [L1] on the effective Palatini variation and pure-pair representative), the flat affine no-go (Corollary 1, [L1]), and the Lorentz-invariant pairing no-go (Theorem 3, [L1]). The pairing escape route is therefore closed for every nonzero nondegenerate symmetric Lorentz-invariant slice pairing. **Remaining lemma, named:** compute the full composite $\Theta$-only variation and construct from $S[\Theta]$ a canonical torsion sector that admits the flat affine representer, either through a derived non-minimal torsion invariant/-cancellation or through a translational or relative-bimodule completion that introduces no independent propagating fields.

**GAP-10D: NARROWED.** The Einstein–$\Lambda$ endpoint remains conditional on the imported $S_{\mathrm{HP}}$ and Lovelock hypotheses (GAP-10D-PALATINI / GAP-10D-UNIQUENESS, unchanged). **Remaining lemma, named (induced-Palatini lemma):** show that the $\Theta$ effective action contains $\epsilon_{abcd}e^a \wedge e^b \wedge R^{cd}(\omega)$ with a sign and coefficient fixed by $(\mathcal{N}_0, \Theta\text{-vacuum})$, e.g. via the curved heat-kernel $a_2$ coefficient of $D^\dagger D$; the present spectral tooling stops at flat tori.

Prohibited conclusions, explicitly not claimed: closure of GAP-10T-DYN or GAP-10D; derivation of $S_{\mathrm{HP}}$; any statement that the composite scheme reproduces Einstein–Hilbert.